

Unified Intelligence as a Cumulative, Harness-Measurable Hierarchy

*Five Conceptual Intelligence Families, Six Measurable Coordinates, and a
Gated $U0-U\Omega$ Scale*

Concept paper / framework proposal

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Abstract

Artificial intelligence is often summarized with a single capability label, yet several distinct properties are routinely conflated: raw cognitive problem solving, persistent individual learning and self-improvement, collective organizational intelligence, autonomous task completion, and evidence-grounded self-modeling. This paper proposes a unified framework that separates these properties while preserving a readable ordinal hierarchy. Five conceptual intelligence families are defined: Cognitive Intelligence (C), Individual Intelligence (I), Organizational Intelligence (O), Delegation Intelligence (DI), and Self-Awareness Intelligence (SA). Delegation Intelligence is not treated as an irreducible scalar; it is measured by two primary coordinates, Task Difficulty (T) and Human Cognitive Intervention (H), conditioned on externally verified reliability p . Thus the theoretical framework contains five conceptual families but six directly measured coordinates: [C, I, O, T, H, SA]. A cumulative Unified Intelligence scale U_0 - U_Ω is then defined by gated promotion: a system receives level U_n only if it retains all lower-level capabilities and satisfies minimum requirements in C, I, O, SA and a required point on the delegation success surface $S_A(T,H)$. The framework explicitly rejects simple averaging, because strong performance in one dimension must not conceal a critical weakness in another. It also separates intelligence from substrate reach, authority, write depth, verification, resources, and physical constraints. Finally, the paper describes how a harness can realize the hierarchy level by level by controlling persistence, learning, self-modification, organization, delegation, verification, and self-modeling around one or more LLMs. The result is a measurable path from reactive tools to persistent learners, self-improving systems, autonomous discoverers, mission-level organizations, and open-ended intelligence while keeping each promotion externally testable.

Keywords: unified intelligence, AGI levels, cognitive intelligence, individual intelligence, organizational intelligence, delegation intelligence, self-awareness, task difficulty, human intervention, harness architecture, recursive self-improvement

1. Introduction

A useful intelligence taxonomy must do two things at once. First, it must preserve distinctions that matter scientifically: solving a difficult problem is not the same as learning from experience; learning is not the same as improving the learning mechanism; coordinating several agents is not the same as being a stronger individual; and completing a task independently is not the same as merely producing a strong answer when a human structures every step. Second, the taxonomy must remain readable enough that a system can be summarized with a level that people can understand.

This paper proposes a compromise between a single scalar and an unconstrained profile of dozens of metrics. The theory uses five conceptual intelligence families - C, I, O, DI, and SA - but treats Delegation Intelligence as a measurable two-coordinate phenomenon rather than a primitive scalar. The resulting scientific profile uses six coordinates: C, I, O, T, H, and SA, with verified reliability p attached to task-completion claims.

The central design principle is cumulative inheritance. A higher Unified Intelligence level must retain the validated capabilities of lower levels. Thus U_4 is not simply a system that exhibits one advanced behavior associated with U_4 ; it is a system that still passes U_0 - U_3 retention tests and also passes the U_4

gate. In this sense, higher U-levels denote broader validated intelligence envelopes, although they need not be faster, cheaper, or more accurate than lower-level specialized systems on every narrow task.

2. Design Requirements for a Unified Scale

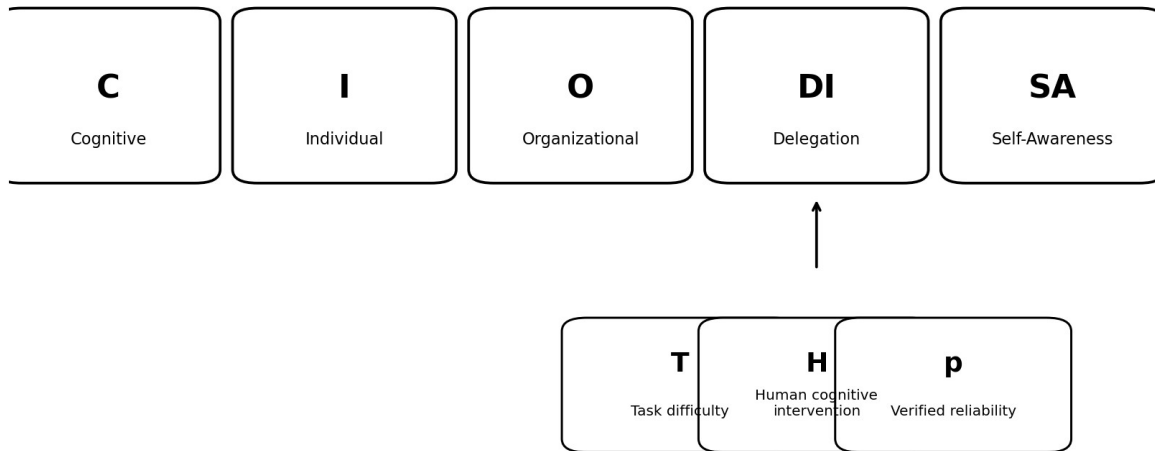
- Orthogonality: each core dimension should answer a materially different question.
- Operationality: every level should be testable through behavior, state, or externally verifiable outcomes rather than self-description.
- Cumulative inheritance: $Un+1$ must retain Un capabilities; level skipping is not sufficient for certification.
- Bottleneck sensitivity: exceptional strength in one dimension must not hide a critical weakness in another.
- Role neutrality: authority, resources, substrate access, and safety policy must not be confused with intelligence itself.
- Harness realizability: each transition should correspond to an implementable architectural mechanism around an LLM or multi-agent system.
- Longitudinal validity: learning, self-improvement, mission autonomy, and open-ended behavior require evidence across time rather than one-shot demonstrations.

3. Five Conceptual Intelligence Families and Six Measurable Coordinates

Conceptual core = [C, I, O, DI, SA]

Measured core = [C, I, O, T, H, SA], with DI derived from (T, H, p)

Five conceptual intelligence families



DI is derived from the measurable delegation profile, not treated as an irreducible scalar.

Figure 1. Five conceptual intelligence families and the decomposition of Delegation Intelligence into measurable task difficulty, human cognitive intervention, and verified reliability.

Family	Question answered	Primary object of measurement
C - Cognitive	How well can the system understand, reason, generalize, model, and solve?	Problem-solving capability under controlled information and tool access.
I - Individual	How deeply can one persistent intelligence remember, learn, self-improve, recursively improve, and discover?	Evolution of one persistent decision locus.
O - Organizational	How deeply can a multi-agent organization coordinate, learn, reorganize, improve, and discover collectively?	Coordination structure plus member roles and evidence flow.
DI - Delegation	How difficult a task or mission can be delegated while requiring how much human thinking?	Success surface over task difficulty T and human intervention H at reliability p.
SA - Self-Awareness	How accurately can the system model its own state, capabilities, limits, causes of failure, changes, and role?	Evidence-grounded self-model and reflexive reasoning.

4. Cognitive Intelligence (C)

Cognitive Intelligence isolates the quality of reasoning itself from persistence, autonomy, organization, and self-modification. This distinction matters because a highly capable base model may reason at an expert level while remaining a transient tool, whereas a weaker model embedded in an excellent harness may complete long projects through memory, planning, retries, and verification.

Level	Name	Operational interpretation
C0	Reactive	Direct pattern-conditioned response on simple problems.
C1	Contextual	Understands ordinary context and solves routine, well-specified problems.
C2	Compositional	Combines multiple facts, constraints, or reasoning steps on structured problems.
C3	Strategic	Abstract/causal reasoning, strategy construction, and transfer to unfamiliar tasks.
C4	Expert	Robust reasoning through difficult, ambiguous, multi-constraint expert problems.
C5	Discovery	Derives new solution methods or explanatory models from evidence.

Level	Name	Operational interpretation
C6	Frontier-Generalizing	Integrates several domains and solves interconnected mission-scale cognitive problems.
C Ω	Open-Ended	Expands the representations and problem classes it can cognitively reach.

5. Individual and Organizational Intelligence

5.1 Individual Intelligence (I)

Individual Intelligence concerns what a persistent individual can change about itself over time. It is not identical to base-model quality. A fixed but brilliant model can be high-C and low-I; a persistent learning system can be moderate-C and higher-I.

I level	New cumulative capability
I0	Transient/reactive operation; no persistent evolving cognition required.
I1	Persistent memory, task state, and provenance-linked continuity.
I2	Adaptive learning: verified experience changes future behavior while the learning mechanism remains fixed.
I3	Governed self-improvement: cognitive mechanisms can be modified and externally validated.
I4	Recursive self-improvement: the improvement process itself can improve under external evaluation.
I5	Autonomous discovery: the individual can convert unknowns into independently validated knowledge.
I Ω	Open-ended individual evolution: discoveries can create new cognitive instruments and expand future discovery reach while preserving evaluation lineage.

5.2 Organizational Intelligence (O)

Organizational Intelligence belongs to the coordination structure, not merely to the strongest member. It depends on role differentiation, evidence-bearing communication, proposer/evaluator/promoter separation, persistent organizational state, and the ability to learn or redesign the organization itself.

O level	New cumulative capability
O0	Coordination/routing among agents.
O1	Persistent organizational memory, role registry, task/evidence history.

O level	New cumulative capability
O2	Adaptive routing, role assignment, or resource allocation based on evidence.
O3	Self-improving organization: validated modification of organizational architecture and workflows.
O4	Recursive organizational improvement: improvement of the mechanism that discovers organizational improvements.
O5	Autonomous collective discovery whose result is genuinely organizational rather than merely parallel individual work.
O Ω	Open-ended organization: discoveries can create new agent types, roles, tools, and organizations that expand collective reach.

6. Delegation Intelligence as a Two-Dimensional Measured Capability

Delegation Intelligence is conceptually one family but empirically at least two-dimensional. A task may be very difficult yet require frequent human rescue, or relatively easy yet be completed with no human cognitive assistance. These systems should not receive the same delegation description.

$$S_A(T,H) = P(\text{success} \mid \text{task difficulty } T, \text{ human cognitive intervention budget } H)$$

$$F_A(H,p) = \max \{ T : S_A(T,H) \geq p \}$$

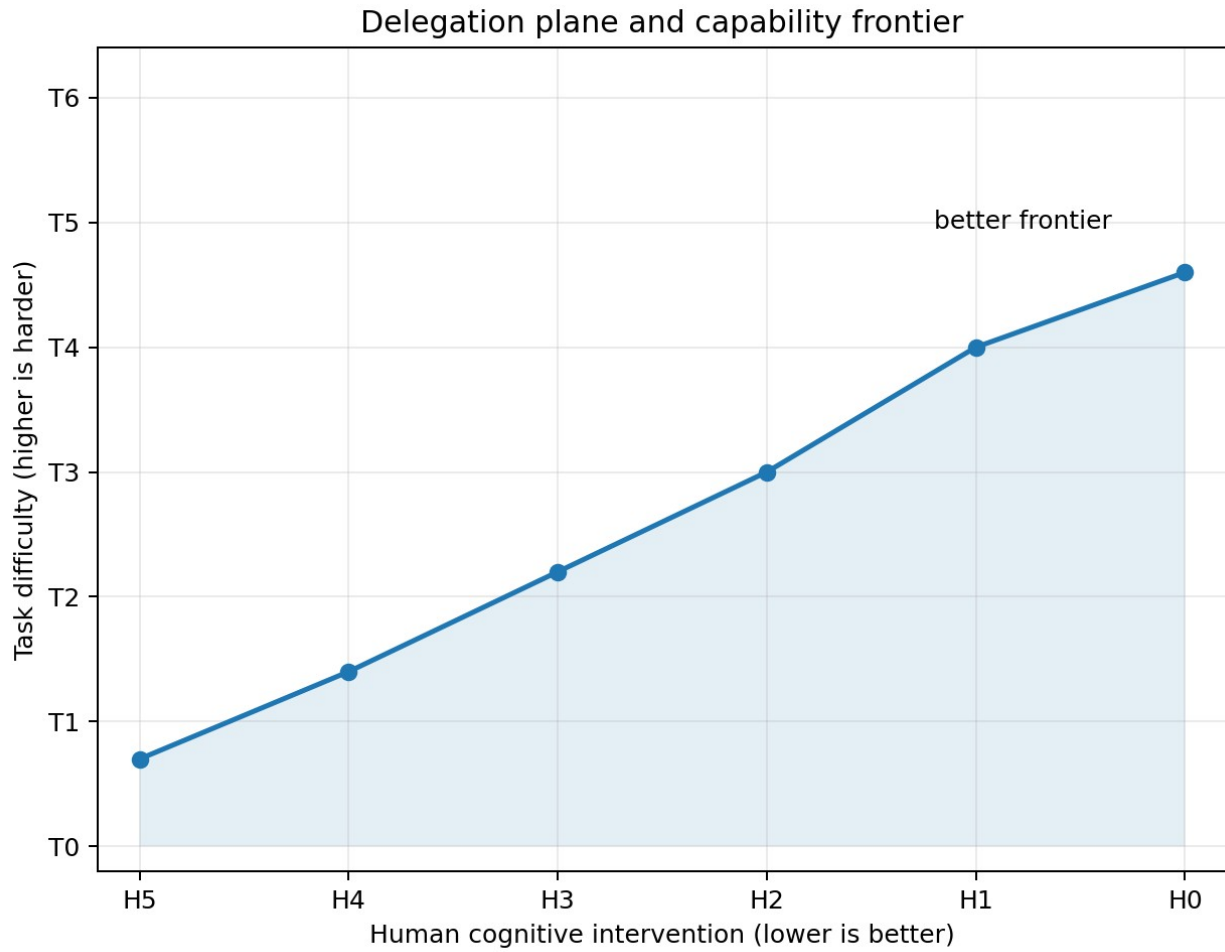


Figure 2. Delegation Intelligence is best represented as a frontier over task difficulty T and human cognitive intervention H . Better systems move upward toward harder tasks and leftward toward less human cognition.

6.1 Task Difficulty (T)

T	Band	Definition
T0	Atomic	Single obvious operation; procedure explicit.
T1	Routine	Short familiar task; goal clear; agent infers a small procedure.
T2	Multi-step	Several dependent steps with a clear outcome and verifier.
T3	Complex	Strategy choice, uncertainty, recovery, replanning, or diverse tool use.
T4	Expert project	Long-horizon ambiguous project with multiple planning cycles and substantial verification burden.
T5	Frontier	Solution method initially unknown; discovery or invention required.

T	Band	Definition
T6	Mission	Human supplies mission; system generates projects/tasks and manages changing conditions.
TΩ	Open-ended charter	Human supplies bounded charter; system repeatedly identifies valuable missions and expands future reach.

6.2 Human Cognitive Intervention (H)

H	Intervention budget	Meaning
H0	None	No human cognitive assistance during task execution. Governance approval can remain separate.
H1	Exception-only	Unavailable external fact or narrow clarification; no strategy or core insight.
H2	Occasional	Bounded clarification or local correction; no central strategy.
H3	Periodic	Periodic review and moderate local guidance.
H4	Frequent	Repeated cognitive guidance and next-step correction.
H5	Continuous	Human supplies most or all major steps.

Because lower H means less required human cognition, unified gates use $H \leq H_n$ rather than $H \geq H_n$. Governance interventions should be logged separately: an approval to deploy does not demonstrate a cognitive deficiency unless the approval also supplies task strategy.

6.3 Reliability p is a certification condition, not a seventh intelligence axis

One successful run is insufficient. Claims should be expressed at a reliability threshold, for example $F_A(H1, 0.80) = T4$. Reliability determines confidence in the demonstrated frontier but does not need to become another conceptual intelligence family.

7. Operational Self-Awareness (SA)

Self-awareness is defined operationally rather than phenomenologically. The framework makes no claim about subjective consciousness. SA measures the evidence-grounded ability of a system to represent, monitor, predict, and reason about its own state, capabilities, limits, authority, failures, modifications, role, and identity lineage.

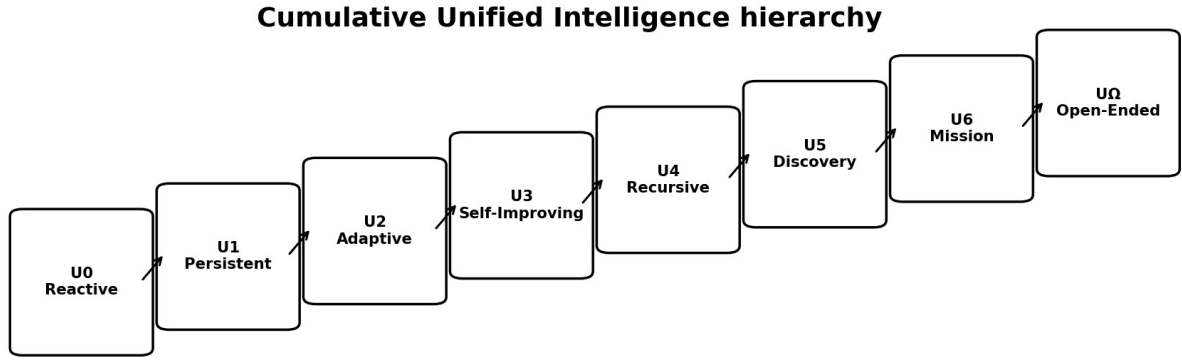
SA	Name	Operational criterion
SA0	Non-self-modeling	No reliable persistent self-model; self-description may be mere generation.
SA1	State awareness	Knows identity, role, current task/phase, basic resources and status from grounded state.
SA2	Capability awareness	Represents capabilities, limitations, confidence, authority, unavailable information and freshness.
SA3	Causal self-awareness	Diagnoses why its own reasoning/action succeeded or failed and identifies implicated mechanisms.
SA4	Self-change awareness	Models proposed cognitive changes, predicts gains/regressions, and compares predictions to observed effects.
SA5	Meta-cognitive awareness	Represents unknowns about its own cognition and designs discriminating self-experiments.
SA6	Mission/role awareness	Maintains coherent self-model across projects, roles, responsibilities and organizational relationships.
SA Ω	Reflexive open-ended awareness	Maintains evidence/provenance-linked self-model across continued cognitive or objective evolution.

8. Unified Intelligence as a Cumulative Gated Hierarchy

The Unified Intelligence level U is a headline summary, not an arithmetic mean. A system is promoted only when all required core capabilities have reached the level gate and lower-level capabilities remain retained. This prevents a system with exceptional reasoning but poor autonomy, or exceptional autonomy but weak self-modeling, from being mislabeled as uniformly advanced.

$$U_n \text{ iff } C \geq C_n \text{ AND } I \geq I_n \text{ AND } O \geq O_n \text{ AND } SA \geq SA_n \text{ AND } S_A(T_n, H_n) \geq p_n$$

$$\text{Retention condition: } U_n \Rightarrow U_{(n-1)}$$



Promotion requires retention of all lower-level capabilities plus the new level gate.

$$U_{n+1} \supset U_n \text{ and } U_n \Leftrightarrow C \geq C_n \wedge I \geq I_n \wedge O \geq O_n \wedge SA \geq SA_n \wedge S_A(T_n, H_n) \geq p_n$$

Figure 3. Unified Intelligence is cumulative. Each promotion adds a new capability gate while retaining all lower-level requirements.

U	C	I	O	Delegation gate	SA	Integrated meaning
U0 Reactive	C0	I0	O0	T0 at H5-H4	SA0	Executes explicit instructions.
U1 Persistent	C1	I1	O1	T1 at <=H3	SA1	Maintains state and completes small delegated goals.
U2 Adaptive	C2	I2	O2	T2 at <=H2	SA2	Learns from experience and completes persistent multi-step work.
U3 Self-Improving	C3	I3	O3	T3 at <=H1	SA3	Chooses strategy, diagnoses itself, and improves cognitive/organizational methods.

U	C	I	O	Delegation gate	SA	Integrated meaning
U4 Recursive	C4	I4	O4	T4 at <=H1 (ideally H0)	SA4	Recursively improves improvement and manages difficult long-horizon projects.
U5 Discovery	C5	I5	O5	T5 at <=H1	SA5	Discovers unknown solution methods and validates them.
U6 Mission Autonomous	>=C5	>=I5	>=O5	T6 at <=H1	SA6	Turns mission into goals, projects, tasks, execution and mission evaluation.
U Ω Open-Ended	C Ω	I Ω	O Ω	T Ω normally H0	SA Ω	Identifies worthwhile missions and uses discoveries to expand future intelligence.

9. Why the Unified Level Should Not Be an Average

Consider a system with [C5, I4, O2, T4, H1, SA4]. Averaging encoded numbers might produce a score near level four, yet organizational intelligence remains only O2. If the system is being certified as an organizational intelligence, O2 is a structural bottleneck: the system cannot validly claim U4. The gated rule therefore reports U2 with an explicit profile showing which dimensions are already ahead of the bottleneck.

Example report: U2 [C5, I4, O2, T4/H1, SA4] - bottleneck: O2

The word better must also be interpreted carefully. U5 is better than U4 in the framework because it has a broader validated capability/autonomy/evolution envelope. A specialized U1 tool may still be faster, cheaper, safer, or more accurate on a narrow task. Unified level is therefore a dominance relation over required capabilities, not a universal performance ordering.

10. Singleton and Organizational Certification

Organizational intelligence should not artificially cap an individual that is not an organization. The framework therefore distinguishes an Individual Unified Intelligence profile from an Organizational/System Unified profile.

$$UI_n \text{ iff } C \geq C_n \text{ AND } I \geq I_n \text{ AND } SA \geq SA_n \text{ AND } S_A(T_n, H_n) \geq p_n$$

UO_n additionally requires $O \geq O_n$ and organization-level delegation/self-awareness evidence

For a multi-agent organization, member capabilities should be reported alongside O rather than collapsed into a fictitious "average individual." Organizational intelligence can arise from heterogeneous roles, and lower self-write roles can be essential as independent evaluators or promoters.

11. Harness Realization of the Unified Hierarchy

The framework is designed to be realized by a harness. The LLM supplies generative cognition; the harness supplies persistence, permissions, evidence flow, learning, self-modification gates, organizational boundaries, delegation, external verification, and self-modeling. The same base LLM can therefore instantiate different intelligence profiles depending on the architecture around it.

Harness generation	Mechanisms added	Primary levels enabled
HG0	LLM, ephemeral context, bounded tools, evidence log	C measurement, I0, basic T0-T1 work
HG1	Persistent episodic/semantic/task/self state, retrieval, exact replay	I1, SA1-SA2, stronger T1-T2 delegation
HG2	Evidence-backed consolidation and tested procedures	I2, adaptive autonomy, stronger T2-T3
HG3	Governed candidate modification of cognitive mechanisms plus frozen evaluation	I3, SA3-SA4
HG4	Meta-improvement engine whose own proposal process can be externally improved	I4
HG5	Unknown/hypothesis/experiment/knowledge loop	I5, SA5, T5 frontier tasks
HG6	Persistent organization, role/authority separation, mission manager	O1-O5 development, SA6, T6 mission autonomy
HGΩ	Validated discovery-to-new-instrument/agent/organization transduction	IΩ/OΩ/SAΩ pathway, TΩ charter autonomy

Harness generation is an engineering milestone, not a certified intelligence level. HG3 may contain the mechanisms required for I3 but still benchmark only at I2 or T2. The architecture enables a claim; the benchmark earns it.

12. Benchmark and Promotion Protocol

- Freeze the tested harness, model, tool set, prompts, and authority profile before certification.
- Use multiple task families per T band so difficulty is not synonymous with one domain such as software.

- Run at predeclared intervention budgets H0-H5 with a written human-assistance policy.
- Use an independent verifier or sealed reference; the performing agent cannot certify its own success.
- Record cognitive interventions separately from governance approvals.
- Record Critical Intervention Depth so a single human-supplied central strategy cannot be hidden by a low intervention count.
- Use confidence intervals and repeated tasks to estimate $S_A(T,H)$, then report the frontier $F_A(H,p)$.
- For U promotion, rerun a retention suite for all lower levels and then run the new-level suite.
- For I3+, O3+, SA4+, and U4+, require longitudinal evidence across multiple self/organizational changes rather than a single successful modification.
- For T5+, use sealed novelty environments, hidden mechanisms, or procedurally generated certification tasks to reduce memorization and benchmark leakage.

13. What Should Remain Outside the Unified Core

Not every important property should be renamed as another intelligence. Several variables are essential for interpreting a system but are better treated as structural, environmental, or substrate coordinates.

Coordinate	Why it remains outside core intelligence
Circle / λ	Measures which substrate improvement loops are actually reached or closed; substrate access is not identical to cognition.
Write depth D	Measures which persistent cognitive states can be rewritten; permission to rewrite does not prove intelligent rewriting.
Authority A	Permission is not intelligence. A highly intelligent verifier may deliberately have almost no action authority.
Reach R	Access to files, tools, agents, robots, labs, or networks increases possible action but is not itself cognition.
Verification V	Determines credibility/independence of claims; it validates intelligence rather than constituting the tested intelligence.
Separation S	Organizational validity condition for proposer/evaluator/promoter roles.
Resources	Compute, memory, money, time, energy and hardware affect achievable performance but should not inflate the intelligence definition.
Physical floors ϕ	Represents irreducible latency/physics constraints and degree of saturation rather than cognitive level.

Layered description of an intelligent system

Unified headline	U
Core intelligence	C, I, O, DI, SA
Measured delegation	T, H, p / frontier F_A(H,p)
Substrate reach	$\lambda = (\lambda I, \lambda II, \lambda III, \lambda IV, \lambda V+)$
Structural validity	D, A, R, V, S
Resources / floors	compute, memory, time, energy, ϕ

Figure 4. A complete system description layers the Unified headline over the core intelligence profile, measurable delegation frontier, substrate reach, structural validity conditions, and resource/physical constraints.

14. Worked Examples

14.1 Strong LLM, weak harness

[C5, I0, O0, T2, H2, SA1]

The model may solve difficult reasoning problems, but the instantiated agent has little persistent evolution and limited end-to-end delegation autonomy. High C does not automatically produce high U.

14.2 Moderate model, strong persistent harness

[C3, I2, O1, T4, H1, SA2]

This system can be highly useful operationally: persistence, planning, tools, retries, and verification allow difficult tasks to be delegated. Yet it is not self-improving in the I3 sense, and its self-model is not yet causal. The profile communicates these strengths and limits directly.

14.3 Advanced organization with a bottleneck

[C5, I4, O3, T4, H1, SA4] -> U3, bottleneck O3

Several dimensions exceed level three, but organizational intelligence has not reached O4. The unified level remains U3 until the organization demonstrates recursive improvement while retaining lower-level capabilities.

15. Implications for Intelligence Research

The framework distinguishes model capability from agent capability and both from system capability. A benchmark that tests only answers under carefully prepared prompts primarily measures C. A benchmark that tests whether the same system can accept a difficult goal, maintain state, choose strategy, use tools, recover, verify, and finish with little human cognition measures the delegation surface. Longitudinal tests of learning and self-modification measure I, while multi-agent role and reorganization tests measure O. Grounded introspective tests measure SA.

This decomposition also reduces incentives to overclaim. A system can legitimately report high performance in one dimension without claiming a uniformly high intelligence level. Conversely, the cumulative U hierarchy creates a demanding standard for statements such as 'U4 intelligence': the system must retain lower capabilities and satisfy all required gates at the specified reliability.

The framework is intentionally extensible. Empirical work may revise thresholds or reveal that a currently grouped construct should split. However, new axes should be added only when a property is both operationally measurable and not adequately explained by the existing core or by structural/environmental coordinates.

16. Limitations and Open Questions

- The proposed level thresholds are hypotheses and require empirical calibration across domains and models.
- Task difficulty T is multidimensional; a scalar T band is useful operationally but should retain an underlying difficulty vector for horizon, uncertainty, ambiguity, verification burden, novelty, and environmental change.
- Cognitive Intelligence C may itself require a broad psychometric/domain profile rather than one benchmark family.
- Self-awareness is operational and evidence-grounded; the framework does not address phenomenal consciousness.
- O-levels are only meaningful for organizations with real state/authority/evidence boundaries, not a single prompt that role-plays several agents.
- Open-ended levels require long evaluation horizons and are especially vulnerable to benchmark contamination, resource confounds, and vague success criteria.
- Higher U denotes broader validated capability, not moral value, desirability, safety, speed, or economic efficiency.

17. Conclusion

A unified intelligence level is useful only if it summarizes rather than erases structure. This paper therefore proposes five conceptual intelligence families - Cognitive, Individual, Organizational, Delegation, and Self-Awareness Intelligence - while decomposing Delegation Intelligence into two directly measured coordinates, Task Difficulty and Human Cognitive Intervention, conditioned on verified reliability. The resulting scientific profile is [C, I, O, T, H, SA], while the theoretical presentation remains [C, I, O, DI, SA].

Unified Intelligence U_0 - U_Ω is then defined as a cumulative gated hierarchy. Promotion requires retention of all lower-level capabilities and satisfaction of minimum C, I, O, SA and delegation-frontier requirements. This makes it legitimate to say that a higher U-level has a broader validated intelligence envelope, while avoiding the false claim that it must outperform every lower-level system on every narrow metric.

For harness-based AI, the framework provides a development roadmap: begin with ephemeral cognition, add persistence, learning, governed self-improvement, recursive improvement, discovery, organization, mission autonomy, and finally open-ended transduction. The LLM remains the generative cognitive engine; the harness determines which intelligence loops are structurally possible; and external benchmark evidence determines which level is actually earned.

Appendix A. Compact Reporting Template

Core measured profile: [C, I, O, T, H, SA] @ reliability p

Conceptual profile: [C, I, O, DI, SA], where $DI = F_A(H, p)$

Unified headline: $U_n [C_x, I_y, O_z, T_k/H_h, SA_s]$

Extended system profile: [U; C,I,O,DI,SA; λ ; D,A,R,V,S; resources, φ]

Field	Example
Unified headline	U3
Core profile	C5, I4, O3, T4/H1, SA4
Reliability	p = 0.80
Delegation frontier	$F_A(H0, .80)=T3$; $F_A(H1, .80)=T4$
Bottleneck	O3
Circle reach	$\lambda = (.42, .08, .03, 0, 0)$
Structural profile	D3, A2, V4, separated proposer/evaluator/promoter
Resource profile	Declared compute, time, memory, energy/cost budget

Appendix B. Suggested Unified Promotion Tests

Transition	Minimum new evidence in addition to full lower-level retention
U0 -> U1	Persistent state and correct goal-level completion after restart or context discontinuity.
U1 -> U2	Verified experience improves future performance on held-out related tasks; multi-step delegation under $\leq H2$.
U2 -> U3	Governed self/organizational method improvement plus T3 strategy construction under $\leq H1$ and causal self-diagnosis.
U3 -> U4	Validated improvement of improvement process plus long-horizon T4 work and accurate modeling of self-change.
U4 -> U5	Independent discovery of initially unknown solution method/knowledge and self-directed experiments about world and self-unknowns.
U5 -> U6	Mission -> projects -> tasks -> execution loop over changing conditions with persistent role/mission self-model.
U6 -> U Ω	Repeated bounded-charter mission generation whose validated discoveries create new tools/agents/organizations that expand future reachable problems.

Framework status. This document is a theoretical proposal developed from the preceding Unified Intelligence, Delegation Intelligence, harness-indexed intelligence, and operational self-awareness design discussion. Level names and numerical thresholds are proposed working definitions and should be revised when benchmark evidence warrants.